

Uniform Transport Følner Sets for Virtually Abelian Groups

Partial solution packet for arXiv:2509.10686

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Status: candidate substantial partial theorem, likely valid. Problem 10.8 has an affirmative answer for every finitely generated virtually abelian group satisfying the source’s no-real-characters hypothesis. Explicit lattice-box sets have transport defect $O(1/N)$. The general amenable-group problem remains open.

1 Source problem and new result

For a finite nonempty $F \subseteq \Gamma$, write

$$u_F = \frac{1}{|F|} \sum_{x \in F} \delta_x.$$

On a finitely generated group with a left-invariant word metric, the Wasserstein distance between finitely supported probabilities equals the Arens–Eells norm of their difference:

$$W_1(\mu, \nu) = \|\mu - \nu\|_{\text{AE}}.$$

Rosendal proves that if Γ is amenable and has no nonzero homomorphism to $\mathbb{R}$, then, for every finite $E \subseteq \Gamma$ and $\varepsilon > 0$, there is a finitely supported probability β with

$$\max_{g, f \in E} W_1(\beta g, \beta f) < \varepsilon \tag{1}$$

[1, Theorem 10.4]. Rational approximation makes β a uniform average of a finite *multiset*. Problem 10.8 asks whether repetitions can always be removed, so that $\beta = u_F$ for a genuine finite set F .

We prove this for the full virtually abelian class.

Idea of the proof

Use every coset of a finite-index lattice subgroup at once. Right translation shifts the lattice box in each coset by a bounded displacement. The sum of those displacements is a homomorphism from Γ to the lattice, so the no-real-characters hypothesis forces its value to vanish. After moving the small boundary defects into one coset layer, the resulting signed lattice kernel has zero mass and zero first moment. Algebraically it is therefore a sum of second differences. A second difference of a normalized box has Arens–Eells norm $O(1/N)$, which is the required transportation estimate. The argument does not provide such sets for general amenable groups.

Theorem 1. *Let Γ be a finitely generated virtually abelian group with no nonzero homomorphism $\Gamma \rightarrow \mathbb{R}$, and equip it with any left-invariant word metric. There are finite nonempty sets $F_N \subseteq \Gamma$ such that, for every $g \in \Gamma$,*

$$W_1(u_{F_N} g, u_{F_N}) = O_g(N^{-1}). \tag{2}$$

From Example 10.7, we see that for the infinite dihedral group the almost translation-invariant averages $\beta \in \Delta D_\infty$ can be taken on the form

$$\beta = \frac{1}{|F|} \sum_{f \in F} f$$

for appropriate finite subsets $F \subseteq D_\infty$. In particular, this means that the Arens–Eells norm $\|\beta - \beta g\|_{\mathcal{A}}$ can be expressed as follows

$$\|\beta - \beta g\|_{\mathcal{A}} = \frac{1}{|F|} \sum_{f \in F \setminus Fg} d(f, \theta(f))$$

for some bijection θ between $F \setminus Fg$ and $Fg \setminus F$. This formulation provides a more obvious generalisation of Følner’s criterion and it is thus natural to wonder if this can always be achieved.

Problem 10.8. *Suppose Γ is a finitely generated amenable group with no non-zero homomorphisms to $\mathbb{R}$ and equip Γ with its left-invariant word metric. Let $\epsilon > 0$ and $E \subseteq \Gamma$ be finite. Can the $\beta \in \Delta\Gamma$ in Theorem 10.4 always be taken to be of the form*

$$\beta = \frac{1}{|F|} \sum_{f \in F} f$$

for some appropriate finite subset $F \subseteq \Gamma$?

Figure 1: The motivation and Problem 10.8 on source page 32. The page was reconstructed from the run’s ingested official arXiv source.

Consequently, for every finite $E \subseteq \Gamma$ and every $\epsilon > 0$, some F_N satisfies

$$\max_{g, f \in E} W_1(u_{F_N} g, u_{F_N} f) < \epsilon.$$

The infinite dihedral example in the source is the rank-one case. The proof below identifies its opposite-drift cancellation as a general transfer identity.

2 Coset displacement has zero total drift

If Γ is finite, take $F_N = \Gamma$. Otherwise choose a normal finite-index torsion-free abelian subgroup $A \cong \mathbb{Z}^d$. Such an A exists after taking the normal core of a torsion-free finite-index abelian subgroup. Fix a set T of representatives, so every $x \in \Gamma$ is written uniquely as

$$x = at, \quad a \in A, \quad t \in T.$$

Write A additively and put $m = |T|$.

For $g \in \Gamma$ and $t \in T$, there are unique $c_t(g) \in A$ and $\sigma_g(t) \in T$ such that

$$tg = c_t(g) \sigma_g(t). \tag{1}$$

Here σ_g is the permutation of the coset space induced by right multiplication.

Lemma 1 (Transfer cancellation). *For every $g \in \Gamma$,*

$$\sum_{t \in T} c_t(g) = 0 \quad \text{in } A.$$

Proof. From (1),

$$c_t(gh) = c_t(g) + c_{\sigma_g(t)}(h).$$

Since σ_g permutes T , the map

$$S(g) = \sum_{t \in T} c_t(g)$$

is a group homomorphism $\Gamma \rightarrow A$. If $S(g) \neq 0$ for some g , a real linear functional on $A \otimes_{\mathbb{Z}} \mathbb{R}$ nonzero at $S(g)$ would make a nonzero homomorphism $\Gamma \rightarrow \mathbb{R}$. This contradicts the hypothesis, so $S = 0$. $\square$

3 Box estimates and layer collapse

Fix a basis $e_1, \dots, e_d$ of A and put

$$B_N = \left\{ \sum_{i=1}^d n_i e_i : 0 \leq n_i < N \right\}, \quad b_N = \frac{1}{|B_N|} \sum_{a \in B_N} \delta_a, \quad F_N = B_N T.$$

For $v \in A$, let τ_v denote translation by v . A coordinatewise boundary count gives

$$\|\tau_v b_N - b_N\|_1 \leq \frac{2 \sum_i |v_i|}{N} \quad (v = \sum_i v_i e_i), \quad (2)$$

with the harmless convention that the right side may exceed 2.

For $s \in T$, write $\iota_s \eta$ for a finitely supported signed measure on the layer As obtained from a signed measure η on A . Fix $t_0 \in T$. If $\eta(A) = 0$, then

$$\|\iota_s \eta - \iota_{t_0} \eta\|_{\text{AE}} \leq d(s, t_0) \|\eta\|_1. \quad (3)$$

Indeed, the left side is represented by $\sum_a \eta(a)(\delta_{as} - \delta_{at_0})$, and left invariance gives $d(as, at_0) = d(s, t_0)$.

For fixed $g \in \Gamma$, set $d_s = c_{\sigma_g^{-1}(s)}(g)$. Right translating F_N and then relabelling the coset layers gives the exact identity

$$u_{F_N} g - u_{F_N} = \frac{1}{m} \sum_{s \in T} \iota_s (\tau_{d_s} b_N - b_N). \quad (4)$$

Each summand has mass zero. Thus (2) and (3) show that replacing every layer s in (4) by the fixed layer t_0 changes the Arens–Eells norm by $O_g(N^{-1})$.

4 The zero-drift kernel is a second difference

Define the finitely supported signed measure

$$\kappa_g = \frac{1}{m} \sum_{s \in T} (\delta_{d_s} - \delta_{t_0}) \quad \text{on } A.$$

It has total mass zero. Lemma 1 also gives zero first moment:

$$\sum_{a \in A} a \kappa_g(a) = \frac{1}{m} \sum_{s \in T} d_s = 0.$$

Lemma 2 (Second-difference decomposition). *If a finitely supported $\kappa \in \mathbb{R}[A]$, $A \cong \mathbb{Z}^d$, has zero mass and zero first moment, then there are finitely many $\alpha_r \in \mathbb{R}$ and $v_r, w_r \in A$ such that*

$$\kappa = \sum_r \alpha_r (\delta_{v_r} - \delta_0) * (\delta_{w_r} - \delta_0).$$

Proof. Let I be the augmentation ideal of the Laurent group algebra $\mathbb{R}[A]$. The zero-mass condition says $\kappa \in I$. Modulo I^2 one has

$$\delta_v - \delta_0 \equiv \sum_{i=1}^d v_i (\delta_{e_i} - \delta_0) \quad (v = \sum_i v_i e_i),$$

also for negative coordinates, by the Laurent identities. Hence the class of κ in I/I^2 is exactly its first-moment vector. That vector is zero, so $\kappa \in I^2$. Since I is generated by the elements $\delta_v - \delta_0$, membership in I^2 gives the asserted finite decomposition. $\square$

For fixed $v, w \in A$, let $\nu_N = \tau_w b_N - b_N$. Pairing each coefficient of ν_N with its translate by v inside the layer At_0 gives

$$\|\iota_{t_0}(\tau_v \nu_N - \nu_N)\|_{\text{AE}} \leq d(vt_0, t_0) \|\nu_N\|_1 = O_w(N^{-1}), \quad (5)$$

where the distance is a fixed constant depending only on v, t_0 and the chosen word metric. Applying Lemma 2 to κ_g and summing (5) yields

$$\|\iota_{t_0}(b_N * \kappa_g)\|_{\text{AE}} = O_g(N^{-1}). \quad (6)$$

5 Proof of the theorem

Proof of Theorem 1. The layer-collapsed form of (4) is exactly $\iota_{t_0}(b_N * \kappa_g)$. The collapse error is $O_g(N^{-1})$ and (6) has the same bound. Therefore $W_1(u_{F_N} g, u_{F_N}) = O_g(N^{-1})$.

For finite E , the triangle inequality gives

$$W_1(u_{F_N} g, u_{F_N} f) \leq W_1(u_{F_N} g, u_{F_N}) + W_1(u_{F_N}, u_{F_N} f),$$

which tends to zero uniformly over the finitely many pairs $g, f \in E$. Choosing N sufficiently large completes the proof. $\square$

6 Verification, scope, and novelty

The proof was independently reconstructed from the source problem. The three points most vulnerable to hidden assumptions were checked explicitly: the cocycle sum is a genuine homomorphism because A is abelian; moving a zero-mass signed boundary measure between coset layers costs its ℓ_1 -mass times a fixed layer distance; and the augmentation-ideal quotient $I/I^2 \cong A \otimes \mathbb{R}$ identifies the remaining obstruction with the first moment.

No computation is used as proof. The crop script was rerun from the local source rendering. A bounded novelty search on 21 August 2026 covered the run indexes, exact title and arXiv-id searches, exact phrases from Problem 10.8, and primary searches combining uniform finite subsets with the Wasserstein/Arens–Eells formulation. No later resolution or statement of this virtually abelian theorem was located.

The result is only partial: it settles the full virtually abelian class under the source hypothesis, but the general finitely generated amenable case and even a virtually nilpotent extension are not proved here.

Human review recommendation. Check identities (4) and (3), the identification of the first moment with the class in I/I^2 , and the second-difference transport bound (5).

References

- [1] C. Rosendal, *Amenability, Optimal Transport and Abstract Ergodic Theorems*, arXiv:2509.10686v1 (2025).